

Multiple Choice

1. **Answer:** A

Options: A) Lipid-soluble antioxidant in cell membranes; B) Fast responses to increase calcium absorption with no change in gene expression; C) Regulation of differentiation of adipocytes; D) Regulation of bone turnover

Note: Correct option: A - Lipid-soluble antioxidant in cell membranes

2. **Answer:** D

Options: A) is defined as the ratio of VO₂ divided by VCO₂; B) decreases on a high carbohydrate diet.; C) increases with fasting.; D) goes beyond a value of 1.0 when exogenous carbohydrate is converted to endogenous fat.

Note: Correct option: D - goes beyond a value of 1.0 when exogenous carbohydrate is converted to endogenous fat.

3. **Answer:** B

Options: A) Synthesis of TMP (thymidine monophosphate); B) Decarboxylation of amino acids to form amine neurotransmitters; C) Synthesis of methionine from homocysteine; D) Carboxylation of pyruvate to oxaloacetate

Note: Correct option: B - Decarboxylation of amino acids to form amine neurotransmitters

4. **Answer:** B

Options: A) Selenophosphate; B) Selenocysteine; C) Selenohistidine; D) Selenate

Note: Correct option: B - Selenocysteine

5. **Answer:** D

Options: A) Glucose neurotoxicity; B) Insulin insensitivity; C) Systemic inflammation; D) All of the above

Note: Correct option: D - All of the above

True / False

6. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

7. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

8. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

9. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'If survival time with the disease increases.'

10. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'May be effected through the epigenetic effects of intrauterine nutritional inadequacy'.

Long Form Response

11. **Answer:** LDL receptor. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

12. **Answer:** Women over age 80 with osteopenia have above average bone mineral density for their age. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key